Project Titan: Cloud Infrastructure & Microservices Specification

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1. System Architecture Overview

Project Titan migrates legacy monolithic services into an event-driven, containerized microservices topology running on Kubernetes. The target service level agreement (SLA) is 99.99% uptime with p99 response latencies below 85 milliseconds across all API endpoints.

2. Core Infrastructure Components

- **API Gateway:** Traefik v3 with mutual TLS (mTLS) termination and token rate limiting. - **Event Streaming:** Apache Kafka multi-broker cluster across three availability zones for cross-region data synchronization. - **Data Persistence:** Polyglot persistence comprising PostgreSQL for transactional integrity, Redis for session and transient cache, and SQLite/Chroma for local edge indexing. - **Authentication & Identity:** Zero-Trust OIDC/OAuth2 authentication with continuous JWT validation and fine-grained Role-Based Access Control (RBAC).

3. Disaster Recovery and Target RTO / RPO

The Recovery Time Objective (RTO) is guaranteed at 15 minutes, and Recovery Point Objective (RPO) is under 60 seconds. Automated hourly database snapshots are cryptographically signed and replicated to geographically isolated air-gapped cold storage vaults.